**26900057** **Reg.No : .....**

**Name : .....**

# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

## MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

## REGULAR EXAMINATION October 2026

## Fifth SEMESTER

## Skill Enhancement Course (SEC) - MG5SECBTS300 - ENTREPRENEURSHIP IN BIOCHEMISTRY

(2024 ADMISSION ONWARDS)

**Duration: 1 Hour 30 Minutes** **Maximum Marks: 50**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

MCQ

Answer all questions

Each question carries **1** marks

1. A Quality Management System (QMS) in a nutraceutical company mainly helps to: [K] / [CO2]

   a). Ensure consistent product quality and compliance

   b). Change product colour frequently

   c). Increase advertising budget

   d). Reduce the number of employees

2. NPD in the nutraceutical industry stands for: [K] / [CO2]

   a). New Product Development

   b). National Policy Document

c). Nutrient Processing Division

d). New Packaging Design

3. Learning from unsuccessful ventures helps entrepreneurs to: [K] / [CO4]

a). Avoid common mistakes b). Avoid innovation

c). Stop product development d). Ignore market conditions

4. What does “USP” stand for in marketing of nutraceuticals? [K] / [CO2]

a). Unique Selling Point

b). Unique Sales Price

c). United Standards Policy

d). Universal Supplement Plan

5. “Shelf life” of a nutraceutical product is defined as: [K] / [CO2]

a). Time taken to manufacture the product

b). Duration for which the product remains safe and effective under specified storage conditions

c). Time taken for delivery to the customer

d). Period of advertising campaign

6. Which of the following is a common lifestyle disorder caused by chronic overeating and low physical activity? [K] / [CO1]

a). Anaemia b). Rickets

c). Scurvy d). Obesity

7. Which class of phytochemicals includes compounds like lycopene and β-carotene? [K] / [CO1]

a). Flavonoids b). Carotenoids

c). Glucosinolates d). Saponins

8. The main function of carbohydrates in the body is to: [K] / [CO1]

a). Form hormones b). Store genetic information

c). Build bones d). Provide energy

9. Which dosage form is LEAST common for dietary supplements? [U] / [CO2]

a). Tablets b). Capsules

c). Injectable solutions d). Powders

10. FDA recognizes three main types of label claims for foods and supplements as: [K] / [CO2]

a). Color, taste, and smell claims

b). Health claims, nutrient content claims, and structure/function claims

c). Color and taste only

d). Shelf life only

**[1x10 = 10]**

## Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Mention two pieces of information that must be correctly displayed along with the net quantity on a label (as per general labeling norms). [K] / [CO2]

12. State two factors that can affect the shelf-life stability of a food product. [U] / [CO2]

13. Name the vitamin obtained from citrus fruits and state its function. [K] / [CO1]

14. What is meant by “rejuvenative herbs” in the context of preventive healthcare? Give one example. [K] / [CO2]

15. What is meant by the interdisciplinary nature of biochemical entrepreneurship? [K] / [CO3]

16. How do herbal supplements support digestion and metabolism? [K] / [CO2]

17. What does GMP stand for, and why is it important in nutraceutical manufacturing? [K] / [CO2]

18. Why is it important to consult a healthcare professional before starting daily supplements? [K] / [CO1]

19. Give two examples of herbal supplements commonly used for stress or immunity support in Indian practice. [K] / [CO2]

20. Give four examples for functional foods. [U] / [CO2]

21. Prepare a balanced diet plan for an eighteen year old individual.Comment on it's significance. [U] / [CO1]

22. Give two differences between the Indian and global scenario regarding the prevalence of lifestyle disorders [K] / [CO1]

**[2x10 = 20]**

## Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Explain the future prospects and growth potential of biochemical entrepreneurship. [U] / [CO4]

24. Elaborate on the benefits of balanced nutrition in promoting physical growth, immune competence, and disease resistance. [K] / [CO1]

25. Explain the contributions of biochemical entrepreneurship to biotechnology and healthcare. [K] / [CO3]

26. How does a lack of iodine affect thyroid function and cause goiter? [K] / [CO1]

27. Explain the common pitfalls that may lead to failure of biochemical ventures. [K] / [CO3]

28. Explain the importance of commercialising biochemical research and innovations. [K] / [CO3]

29. Explain the emerging trends in biochemical entrepreneurship. [K] / [CO4]

**[4x5 = 20]**